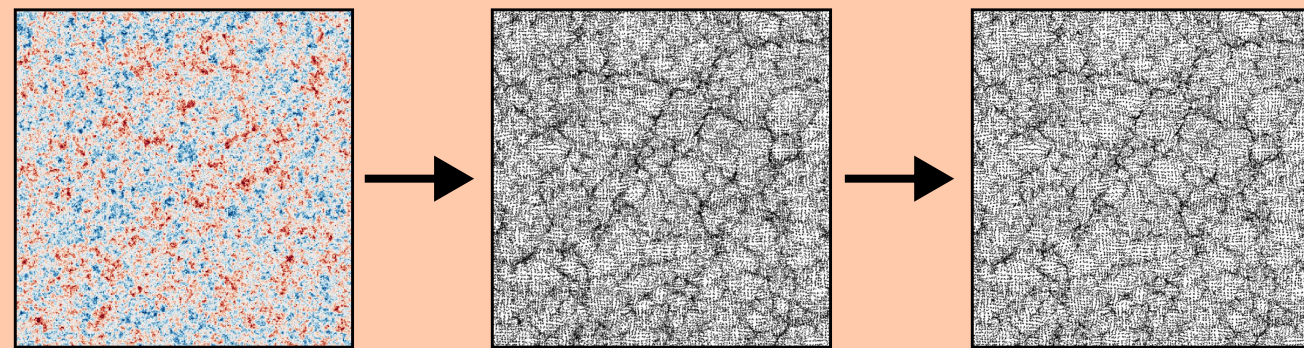


muchisimocks forward model

emulated N-body simulations

initial conditions Zel'dovich approximation emulated displacements

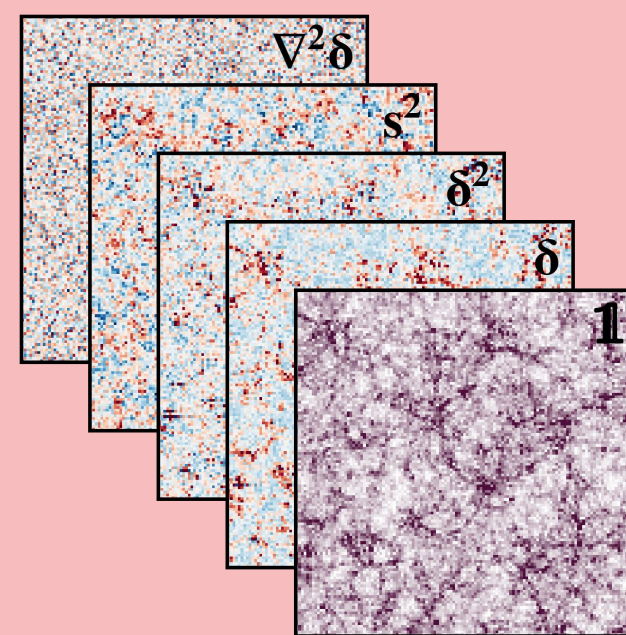


$\uparrow$
 $\{\theta_{\text{cosmo}}\}$

x 10,000 cosmologies

hybrid Lagrangian bias expansion

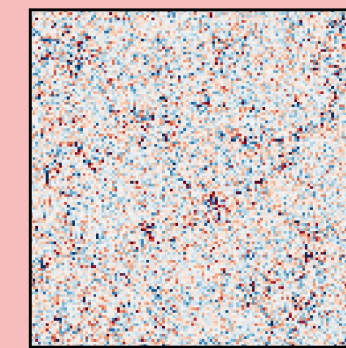
Eulerian bias fields



$\rightarrow$

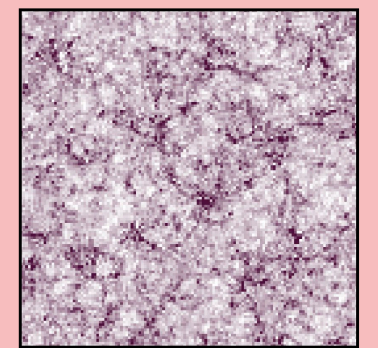
$\downarrow$
 $\{\theta_{\text{bias}}, \theta_{\text{noise}}\}$

noise field



$\rightarrow$

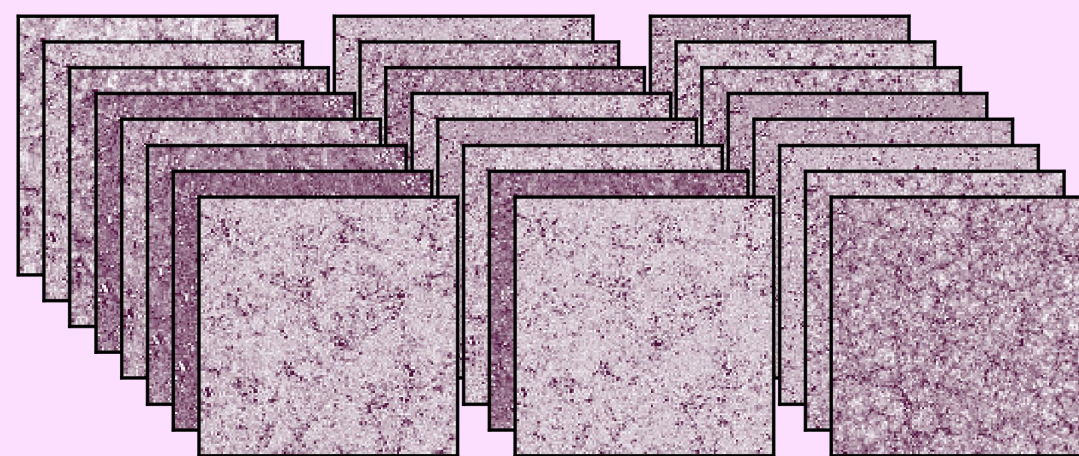
tracer field



x 32 bias+noise parameter sets

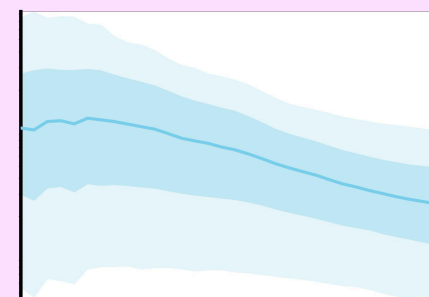
simulation-based inference procedure

training data from
forward model



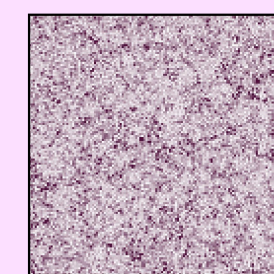
$\downarrow$
summary statistics
 $x \sim p(x|\theta)$

prior
 $\theta \sim p(\theta)$

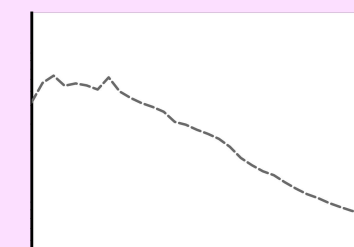


neural density
estimator
 $q_{\phi}(\theta|x)$

test data
deposited to field



$\downarrow$
summary statistics $x_{\star}$



posterior
 $p(\theta|x_{\star})$

